

A pre-feasibility study for the Establishment of Plastic Containers Factory by Using Polyethylene in Jerash



2017



Table of contents

1.	EXECUTIVE SUMMARY	7
	KEY HIGHLIGHT OF JORDAN.....	7
	JORDAN INVESTMENT COMMISSION AIMS TO:	12
2.	MARKET ANALYSIS.....	13
	<i>Project Description</i>	<i>13</i>
	<i>Project Objectives.....</i>	<i>13</i>
	<i>Proposed Product</i>	<i>13</i>
	<i>Target Segments.....</i>	<i>13</i>
	<i>Market Size Analysis</i>	<i>13</i>
	<i>Competitors Analysis</i>	<i>14</i>
	<i>Price Analysis and Pricing Policy.....</i>	<i>14</i>
	<i>Market Share.....</i>	<i>14</i>
	<i>Projected Revenues.....</i>	<i>15</i>
3.	TECHNICAL ANALYSIS	15
	<i>Required Human Resources</i>	<i>15</i>
	<i>Technical Requirements Analysis</i>	<i>17</i>
	<i>1.1.1 Land and construction.....</i>	<i>17</i>
	<i>Machinery and Equipment.....</i>	<i>18</i>
	<i>Vehicles and Transportation</i>	<i>18</i>
	<i>Production Process Description.....</i>	<i>18</i>
4.	FINANCIAL ANALYSIS.....	19
	<i>Assumptions.....</i>	<i>21</i>
	<i>Capital Expenditures</i>	<i>23</i>
	<i>Operating Expenses.....</i>	<i>26</i>
	<i>General and Administrative Expenses</i>	<i>26</i>
	<i>Marketing Expenses</i>	<i>27</i>
	<i>Human Resources.....</i>	<i>27</i>
	<i>Depreciations.....</i>	<i>30</i>

<i>Loan.....</i>	<i>31</i>
<i>Income Statement.....</i>	<i>33</i>
<i>Expected Cashflows Statement.....</i>	<i>34</i>
<i>Expected Balance Sheet.....</i>	<i>35</i>
<i>Feasibility Indicators</i>	<i>37</i>
<i>Sensitivity Analysis</i>	<i>38</i>

List of Tables

Table 1 economical indicators	7
Table 2 Population Distribution according to Age Group.....	8
Table 3 Jordan production of fruits (tons)	13
Table 4 Market Share.....	14
Table 5 project revenues during the years 2018 to 2027	15
Table 6 analysis of production capacity	15
Table 7 The required labor	15
Table 8 the required land for the project.....	17
Table 9 Construction Works	18
Table 10 Machinery and Equipments	18
Table 11 Vehicles and Transportation.....	18
Table 12 General Assumptions	21
Table 13 Currency Exchange Rates	21
Table 14 Annual Growth Rates Assumptions	21
Table 15 Expenses Assumptions.....	21
Table 16 Income Tax Assumptions	21
Table 17 : Weighted Average Cost of Capital	22
Table 18 Land Capital Expenditure.....	23
Table 19 Construction Work Capital Expenditure	23
Table 20 Machinery and Equipment Capital Expenditure	23
Table 21 Furniture Capital Expenditure	23
Table 22 Transportation and Vehicles Capex	23
Table 23 Pre Operating Expenses.....	24
Table 24 Working Capital	24
Table 25 Expenses Summary.....	24
Table 26 Sources of Funding	25
Table 27 Operating Expenses	26
Table 28 General and Administrative Expenses	26
Table 29 Marketing Expenses	27
Table 30 Expected staff preparation.....	28
Table 31 Expected monthly salaries (2018-2027)	29

Table 32 Capex and Depreciation Expenses	31
Table 33 Depreciations and Additions on Construction Works	31
Table 34 Loan Details	32
Table 35 Income Statement.....	33
Table 36 Expected Cashflows	34
Table 37 Expected Balance Sheet.....	36
Table 38 Free Net Cash flows Table	37
Table 39 payback period	38
Table 40 Financial Analysis Results.....	38
Table 41 Sensitivity Analysis	39

Table of Figures

Figure 1 Population of Jordan.....	8
Figure 2 Sectors Contribution to Jordan's Economy.....	9

1. Executive Summary

Polyethylene is a product classified under the same thermal plastic products. Each year more than 60 million tons are being produced worldwide, and this material was included in producing plastic containers are used to put eggs and fruits in them.

The project aims to establish a factory to manufacture plastic containers of polyethylene, which is imported and the increasing demand in the domestic market is noticed. The project will be established in Jerash Governorate.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	487,266
Internal Rate of Return (IRR)	25.68%
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	291,216
Job Opportunities provided by the Project	28
Total Revenues (For 10 Years)	5,187,500
Total Operating Expenses (For 10 years)	2,415,744
Total Net Profit (For 10 years)	690,650

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

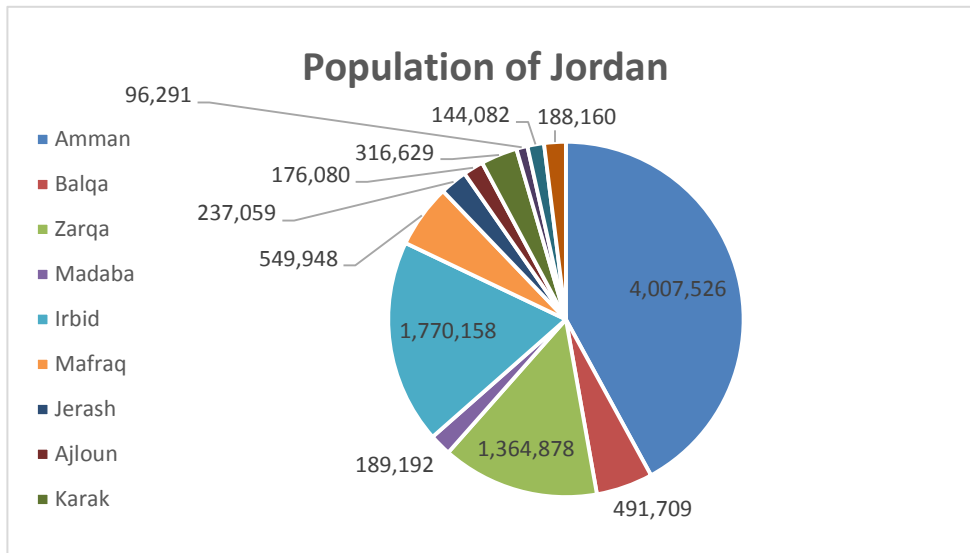
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

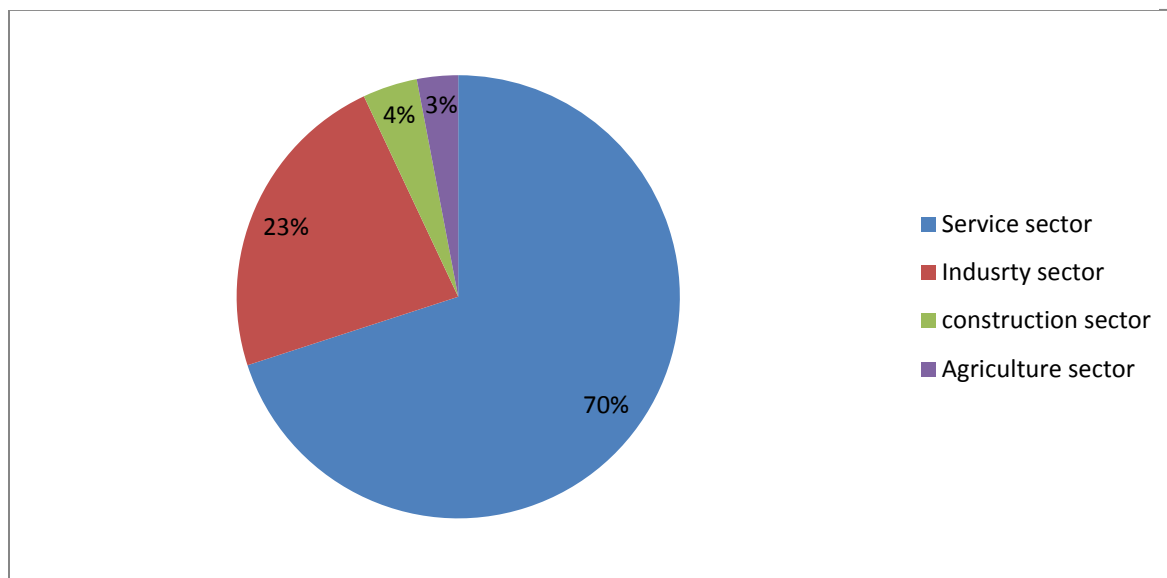
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the

Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally. The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and

legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.

- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

2. Market Analysis

Project Description

This study aims to establish a factory in Jerash to manufacture plastic containers for storing fruits and eggs by using polyethylene as a raw material, where the advantages of polyethylene are its high-density, very light and excellent transparency, as well as its strength and resistance to breakage and ability for recycling.

Project Objectives

The projects aims to achieve the following:

- Produce environmental friendly plastic containers to be recycled.
- Provide job opportunities.
- Achieve financial return for the owners.

Proposed Product

The proposed project provides containers made of plastic with a thickness ranging from 0.25 -1 millimeter made of polyethylene for storing fruits, vegetables and eggs.

Target Segments

Targeted segments for this project are fruit, eggs and plastic containers traders.

Market Size Analysis

Egg consumption in 2016 averaged about 60 million eggs per month, 20% of which are packed in plastic boxes, with the assumption that the plastic enclosure capacity is 20 then the market size of plastic containers for eggs equals 600,000 packet / month.

The following table shows Jordan production (tons) and Jerash production of fruits that are stored inside polyethylene containers:

Table 3 Jordan production of fruits (tons)

	Pears	almonds	peach	apricot	cherries	Figs	Palm	Strawberries
Jordan	9112.259	9733.1	58778.85	28785	4665.3	6911.35	2203.65	16671

Jerash	0	312	36	842	23	479	0	0
--------	---	-----	----	-----	----	-----	---	---

Annual Statistical Report for 2015, the Ministry of Agriculture / Department of Policy Studies / Department statistics and agricultural data

Jordan's imports of fruit (kiwi) reached 3827.8 tons per year. Thus, the total annual market is 11,189,000 packet of eggs and fruits.

Competitors Analysis

The market analysis found that Ammon factory is the only factory that makes plastic containers of polyethylene, with high density.

Price Analysis and Pricing Policy

The price per ton of polyethylene equals to 1400 dinars including taxes and will be sold per carton, where one carton contains 100-400 packet, the price of one carton ranging from 12.5-50 dinars

Market Share

The production capacity of the project will be 5,000,000 packet per year, and the selling will be based on the following percentages:

Table 4 Market Share

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production rate	65%	70%	75%	75%	80%	80%	80%	85%	85%	85%

Projected Revenues

The following table shows the projected revenues for 10 years from the start date of the project:

Table 5 project revenues during the years 2018 to 2027

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production rate	70%	70%	80%	80%	85%	85%	85%	90%	90%	95%
Total revenue	455000	455000	520000	520000	552500	552500	552500	585000	585000	617500

The projected annual revenues for the project based on the expenses of the analysis of production capacity of the project as per the following table:

Table 6 analysis of production capacity

Analysis of Productivity	
Annual production capacity (pack)	5000000
The size of the local market	11189000
Selling price	0.13

3. Technical analysis

Required Human Resources

Table 7 The required labor

Job title	No.
Indirect employees	
Project Manager	1

Job title	No.
Financial Manager	1
Sales and Marketing Manager	1
Marketing team	2
Accountant	2
Guard	1
Human Resources Manager	1
Human Resources staff	1
Total	10
Direct employees	
Production Manager	1
Production Supervisor	1
Production employees	12
Maintenance employees	2
Driver	2
Grand total	28

Job Description:

General Manager: will perform all the tasks and responsibilities of the general manager, and have active participation in the identification and formulation of objectives, as well as planning and organizing workflow of the facility to ensure the achievement of specific objectives.

Sales Manager: will be responsible for the planning and organization of the sales and control over the sales activities and also be responsible for identifying the needs of the department of human resources and their qualifications and needs.

Marketing Manager: will plan, direct, coordinate and control the marketing of products.

Marketing Staff: will perform the required business marketing according to the provided plan and approved by the director of marketing and sales in order to achieve the best possible reach of products.

Accountant: will be responsible for the implementation and writing financial and accounting procedures in the plant.

Guard: will maintain security according to the factory security instructions.

Production Manager: oversees the production, packaging, machinery, maintenance, industrial processes, item design and operations expansion.

Production Supervisor: will be responsible for the follow-up and implementation of the production plan, ensure product quality and alignment with the required specifications, supervising the workers and train them and provide daily reports on the production process.

Production employees: will help technicians do some work, working on machinery, packaging materials and carrying out transport and handling work.

Maintenance employees: will keep the machines working and provide periodic maintenance and repair when broken.

Technical Requirements Analysis

1.1.1 Land and construction

The project will be constructed on an area of 4000 sq.m in Jerash Governorate. The following table shows the needed land:

Table8 the required land for the project

Land	Area sq.m
The required land	4,000

The following table shows the required construction work for the project:

Table 9 Construction Works

Construction work	Area (m ²)
Construction of Hangar	1,000
Building works	150
Paving of land	100

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 10 Machinery and Equipments

Machinery and equipment	Quantity
Cutting and forming machine	2
Rolling machine	2
Molds	2
Fire-fighting equipment	1
Air Conditioners	5

Vehicles and Transportation

The following table shows a list of vehicles and means of transportation required for the project

Table 11 Vehicles and Transportation

Vehicles and Transportation	Quantity
Truck	2

Production Process Description

1) Raw material of (polyethylene) is imported from Saudi Arabia (SABIC)

- 2) The first machine converts raw material to rolls since its production capacity is: 30-350 meters/ minute
- 3) The second machine forms the rolls
- 4) After forming the containers, they will be cut using a knife
- 5) With a production capacity of 30 cycles / minute
- 6) Filling of 250-400 packet per carton.



4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.67% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.

- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 12 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	8.00%
Utilities Expenses from Total Revenues	4.00%
Maintenance Expenses from Total Revenues	1.00%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	1.00%

Table 16 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	20%
Income Tax Deduction(2)	60%
Income Tax after Deduction	8%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premium	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	9.48%
Market Risk Premium (3)	10.12%
Income Tax Rate	8.00%
Beta (4)	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 17 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	8%
Bets	1.00
Equity	243,633
Loans Value	243,633
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	243,633
Loans Percentage	%50
Equity	
Cost of Equity	%16.6
Equity Value	243,633
Equity Percentage	%50
Gross	
Project Value	487,266
Proportion of the project	%100
Weighted Average Cost of Capital	12.67%

Capital Expenditures

The estimated cost of the project is 487,266 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 18 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square)	Cost	Area (M2)	Cost
Required Land	7		4,000	28,000
Cost				28,000

Table 19 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Construction of Hangar	100	1,000	100,000
Building works	250	150	37,500
Paving of land	12	100	1,200
Other works	-	-	10,000
Total			148,700

Table 20 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost / JD	Total Cost
Cutting and forming machine	1	500	500
Rolling machine	5	750	3,750
Molds	2	20,000	40,000
Fire-fighting equipment	2	25,000	50,000
Air Conditioners	2	7,500	15,000
Total (JD)			109,250

Table 21 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Furniture	-	-	12,000
Total			12,000

Table 22 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total
----------------------------	----------	---------------	-------

			Cost/JOD
Truck	2	25,000	50,000
Total			50,000

Table 23 Pre Operating Expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	2,000
Transportation	1,500
Marketing	2,500
Training	1,000
Total	7,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	53,584
General and administrative expenses	3	%25	2,341
Marketing expenses	3	%25	1,094
Indirect Salaries	3	%25	31,000
Total			88,019

Table 25 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	28,000	10%	30,800	%6.3
Construction works	148,700	10%	163,570	%33.6
Machinery & Equipment	109,250	10%	120,175	%24.7
Furniture	12,000	10%	13,200	%2.7
Vehicles	50,000	10%	55,000	%11.3
Pre-Operating Expenses	7,000	10%	7,700	%1.6
Working capital	88,019	10%	96,821	%19.9
Total (JD)	442,969		487,266	100%

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (487,266) JD, where 50% of the project will be financed through loans (243,633) JD, and the remaining (50%) through (Equity) with (243,633) JD. The following table summarizes the general structure of the required financing:

Table 26 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	243,633	%50
Loans	243,633	%50
Total	487,266	%100
Use of Fund	Value	Ratio
Capital Expenditures	382,745	%78.55
Pre-Operating Expenses	7,700	%1.58
Working Capital	96,821	%19.87
Total	487,266	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 27 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials	35,000	35,000	40,000	40,000	42,500	42,500	42,500	45,000	45,000	47,500
Utilities	17,500	17,500	20,000	20,000	21,250	21,250	21,250	22,500	22,500	23,750
Maintenance	4,375	4,375	5,000	5,000	5,313	5,313	5,313	5,625	5,625	5,938
Disposables	4,375	4,375	5,000	5,000	5,313	5,313	5,313	5,625	5,625	5,938
Direct Salaries	133,600	125,753	130,909	136,276	141,863	147,680	153,734	160,038	166,599	173,430
Others	19,485	18,700	20,091	20,628	21,624	22,205	22,811	23,879	24,535	25,655
Total	214,335	205,703	221,000	226,904	237,862	244,260	250,920	262,666	269,884	282,210

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 28 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,000	2,060	2,122	2,185	2,251	2,319	2,388	2,460	2,534	2,610
Hospitality	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Stationary and Printing	600	618	637	656	675	696	716	738	760	783

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Transportaton	1,800	1,854	1,910	1,967	2,026	2,087	2,149	2,214	2,280	2,349
Training	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Insurance	1,914	1,971	2,030	2,091	2,154	2,219	2,285	2,354	2,424	2,497
Miscellaneous	851	877	903	930	958	987	1,017	1,047	1,078	1,111
Total	9,365	9,646	9,935	10,233	10,540	10,857	11,182	11,518	11,863	12,219

Marketing Expenses

Table 29 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	4,375	4,375	5,000	5,000	5,313	5,313	5,313	5,625	5,625	5,938

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 30 Expected staff preparation

Job title	No.	Monthly salaries	Total monthly salaries	Total yearly salaries
Project manager	1	2,500	2,500	40,000
Financial Manager	1	1,500	1,500	24,000
Sales and Marketing Manager	1	1,750	1,750	28,000
Marketing team	1	1,500	1,500	24,000
Accountant	1	1,250	1,250	20,000
Guard	1	450	450	7,200
Human Resource Manager	2	450	900	14,400
Human resources staff	2	450	900	14,400
Production Manager	1	450	450	7,200
Production Supervisor	1	400	400	6,400
Production employees	1	380	380	6,080
Maintenance employees	13		9,850	185,600
Driver	3	400	1,200	19,200
Grand Total	30		16,050	284,800

Table 31 Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
Project manager	2,000	2,082	2,167	2,256	2,349	2,445	2,545	2,650	2,758	2,871
Financial Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Sales and Marketing Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Marketing team	450	468	488	508	528	550	573	596	621	646
Accountant	500	521	542	564	587	611	636	662	690	718
Guard	350	364	379	395	411	428	445	464	483	502
Human Resource Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Human resources staff	500	521	542	564	587	611	636	662	690	718
Direct staff										
Production Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Production Supervisor	750	781	813	846	881	917	954	994	1,034	1,077
Production employees	400	416	433	451	470	489	509	530	552	574
Maintenance employees	400	416	433	451	470	489	509	530	552	574
Driver	450	468	488	508	528	550	573	596	621	646

Table 31 Annual Expected Salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
-----------	------	------	------	------	------	------	------	------	------	------

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
Project manager	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Financial Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Sales and Marketing Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Marketing team	14,400	14,990	15,605	16,245	16,911	17,604	18,326	19,077	19,860	20,674
Accountant	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
guard	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Human Resource Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Human resources staff	8,000	8,328	8,669	9,025	9,395	9,780	10,181	10,599	11,033	11,485
Total Indirect	124,000	129,084	134,376	139,886	145,621	151,592	157,807	164,277	171,012	178,024
Direct staff										
Production Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Production Supervisor	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Production employees	76,800	66,624	69,356	72,199	75,159	78,241	81,449	84,788	88,264	91,883
Maintenance employees	12,800	13,325	13,871	14,440	15,032	15,648	16,290	16,958	17,653	18,377
Driver	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Total Direct	133,600	125,753	130,909	136,276	141,863	147,680	153,734	160,038	166,599	173,430
Grand Total	257,600	254,837	265,285	276,162	287,484	299,271	311,541	324,315	337,612	351,454
Depreciations										

The following tables illustrate cost of Capex and Depreciation rates:

Table 32 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	30,800	%0.0	%0.0
Construction Works	163,570	%5.0	%0.0
Machinaries	120,175	%10.0	%2.0
Furniture	13,200	%10.0	%5.0
Vehicles	55,000	%10.0	%0.0
	382,745		

Table 33 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	382,745	385,809	388,953	392,181	395,496	398,900	402,396	405,987	409,677	413,468
Total Depreciation Expenses	27,016	27,322	27,637	27,960	28,291	28,631	28,981	29,340	29,709	30,088
Total Accumulated Depreciation	27,016	54,338	81,975	109,935	138,226	166,857	195,838	225,179	254,888	284,976
Total Additions	0	3,064	3,145	3,228	3,315	3,404	3,496	3,591	3,690	3,791
Total Net Book Values	355,729	331,470	306,978	282,247	257,270	232,042	206,557	180,808	154,789	128,492

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 34 Loan Details

Loan Value	243,633	865,952
Annual Interest Rate	9.48%	9.48%
Loans Period	5	5
Loan Starts at year:	2018	865,952
Annual Payment	63,418	225,409
Number of Payments	5	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				243,633
2019	63,418	23,096	40,322	203,311
2020	63,418	19,274	44,144	159,167
2021	63,418	15,089	48,329	110,838
2022	63,418	10,507	52,911	57,927
2023	63,418	5,491	57,927	0

Income Statement

Table 35 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	437,500	437,500	500,000	500,000	531,250	531,250	531,250	562,500	562,500	593,750
Gross Operating Revenues	437,500	437,500	500,000	500,000	531,250	531,250	531,250	562,500	562,500	593,750
Operating Expenses	-214,335	-205,703	-221,000	-226,904	-237,862	-244,260	-250,920	-262,666	-269,884	-282,210
Gross Operating Profit	223,165	231,797	279,000	273,096	293,388	286,990	280,330	299,834	292,616	311,540
<i>Gross Profit Percentage</i>	<i>%51</i>	<i>%53</i>	<i>%56</i>	<i>%55</i>	<i>%55</i>	<i>%54</i>	<i>%53</i>	<i>%53</i>	<i>%52</i>	<i>%52</i>
Salaries and Benefits (Indirect Staff)	-124,000	-129,084	-134,376	-139,886	-145,621	-151,592	-157,807	-164,277	-171,012	-178,024
General and Administraive Expenses	-9,365	-9,646	-9,935	-10,233	-10,540	-10,857	-11,182	-11,518	-11,863	-12,219
Markeing Expenses	-4,375	-4,375	-5,000	-5,000	-5,313	-5,313	-5,313	-5,625	-5,625	-5,938
Pre Operating Expenses	-7,700									
Gross Indirect Expenses	-145,440	-143,105	-149,312	-155,119	-161,474	-167,761	-174,302	-181,420	-188,501	-196,181
Income before Interest, Depreciation and Tax	77,725	88,692	129,689	117,977	131,914	119,229	106,028	118,414	104,115	115,359
Fixed Assets Depreciations	-27,016	-27,322	-27,637	-27,960	-28,291	-28,631	-28,981	-29,340	-29,709	-30,088
Income before Tax and Interests	50,709	61,370	102,052	90,017	103,623	90,598	77,047	89,074	74,406	85,271
Bank Interests	-23,096	-19,274	-15,089	-10,507	-5,491	-0	-0	-0	-0	-0
Income Before Tax	27,613	42,096	86,963	79,510	98,131	90,598	77,047	89,074	74,406	85,271
Income Tax	-2,209	-3,368	-6,957	-6,361	-7,850	-7,248	-6,164	-7,126	-5,952	-6,822

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	25,404	38,728	80,006	73,149	90,281	83,350	70,883	81,948	68,454	78,449
Net Profit Percentage	%6	%9	%16	%15	%17	%16	%13	%15	%12	%13
Compulsory Reserves	-2,540	-3,873	-8,001	-7,315	-9,028	-8,335	-7,088	-8,195	-6,845	-7,845
Retained Earnings	22,863	57,718	129,724	195,558	276,811	351,825	415,620	489,373	550,981	621,585

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 36 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	25,404	38,728	80,006	73,149	90,281	83,350	70,883	81,948	68,454	78,449
Non-monetary adjustments										
Bank benefits	23,096	19,274	15,089	10,507	5,491	0	0	0	0	0
Consumption	27,016	27,322	27,637	27,960	28,291	28,631	28,981	29,340	29,709	30,088
Operating cash flows before changes in working capital	75,516	85,324	122,732	111,616	124,063	111,981	99,864	111,288	98,163	108,537
Shortage (increase) in inventory	-2,917	0	-417	0	-208	0	0	-208	0	-208
(Decrease) increase in accounts payable	44,834	-2,158	2,574	1,476	2,115	1,600	1,665	2,311	1,804	2,457
Change in working capital	41,917	-2,158	2,157	1,476	1,906	1,600	1,665	2,103	1,804	2,248

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net cash flows (used) from operating activities	117,433	83,166	124,889	113,092	125,970	113,581	101,529	113,391	99,967	110,786
Cash flows from investing activities										
(Purchase) of fixed assets	-382,745	-3,064	-3,145	-3,228	-3,315	-3,404	-3,496	-3,591	-3,690	-3,791
Net flows (used) from investment activities	-382,745	-3,064	-3,145	-3,228	-3,315	-3,404	-3,496	-3,591	-3,690	-3,791
Cash flows from financing activities										
Equity capital	243,633									
Loan Amortization	-40,322	-44,144	-48,329	-52,911	-57,927	0	0	0	0	0
Bank Interest Rate	-23,096	-19,274	-15,089	-10,507	-5,491	-0	-0	-0	-0	-0
Loans	243,633	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	423,847	-63,418	-63,418	-63,418	-63,418	0	0	0	0	0
Net increase (decrease) in cash	158,535	16,685	58,326	46,446	59,237	110,177	98,033	109,800	96,278	106,994
Cashflows at the Beginning of Period	0	158,535	175,220	233,546	279,992	339,229	449,406	547,439	657,239	753,516
Cashflows at the End of Period	158,535	175,220	233,546	279,992	339,229	449,406	547,439	657,239	753,516	860,511

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 37 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	158,535	175,220	233,546	279,992	339,229	449,406	547,439	657,239	753,516	860,511
Inventory	2,917	2,917	3,333	3,333	3,542	3,542	3,542	3,750	3,750	3,958
Total assets traded	161,452	178,137	236,880	283,325	342,770	452,947	550,980	660,989	757,266	864,469
Non-current assets										
Fixed assets (net)	355,729	331,470	306,978	282,247	257,270	232,042	206,557	180,808	154,789	128,492
Total non-current assets	355,729	331,470	306,978	282,247	257,270	232,042	206,557	180,808	154,789	128,492
Total assets	517,181	509,607	543,857	565,572	600,041	684,990	757,538	841,797	912,055	992,961
Liabilities										
Current liabilities										
Payables	44,834	42,676	45,250	46,726	48,841	50,440	52,105	54,417	56,221	58,678
Remaining amount of Loan	44,144	48,329	52,911	57,927	0-	0-	0-	0-	0-	0
Total current liabilities	88,978	91,005	98,161	104,653	48,841	50,440	52,105	54,417	56,221	58,678
Non-current liabilities										
long term loans	159,167	110,838	57,927	0	0	0	0	0	0	0
Total long - term liabilities	159,167	110,838	57,927	0	0	0	0	0	0	0
Total liabilities	248,145	201,842	156,087	104,653	48,841	50,440	52,105	54,417	56,221	58,678
Property rights										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Member Contributions	243,633	243,633	243,633	243,633	243,633	243,633	243,633	243,633	243,633	243,633
Compulsory reserve	2,540	6,413	14,414	21,729	30,757	39,092	46,180	54,375	61,220	69,065
Profit (loss)	22,863	57,718	129,724	195,558	276,811	351,825	415,620	489,373	550,981	621,585
Total equity	269,036	307,764	387,770	460,919	551,200	634,550	705,433	787,380	855,834	934,283
Total liabilities and equity	517,181	509,607	543,857	565,572	600,041	684,990	757,538	841,797	912,055	992,961

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 38 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-487,266	125,133	80,103	121,744	109,864	122,655	110,177	98,033	109,800	96,278	106,994	1,283,321
Discount Factor	1.00	0.70	0.62	0.55	0.49	0.43	0.39	0.34	0.30	0.27	0.24	0.24
Net Present Value for Cash	-487,266	87,486	49,705	67,049	53,702	53,211	42,423	33,502	33,303	25,918	25,564	306,619

Table below (39) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 39 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-487,266	125,133	80,103	121,744	109,864	122,655	110,177	98,033	109,800	96,278	106,994
Project Value Returned	487,266	362,133	282,030	160,285	50,421	-72,233	-182,410	-280,443	-390,243	-486,521	-593,515
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 40 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.67
Net Present Value for Cashflows	291,216
Payback Period	4
Internal Rate of Return	%25.68

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 294,216 Jordan Dinars considering that the project provides 28 Job Opportunities for the governorate residents.

Table 41 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%25.68	4	12.67%
Revenues declined by 10%	%15.98	6	12.67%
Operating Expenses Increased by 10%	%20.55	5	12.67%